

Final Project Report

Date	01 October 2026
Team ID	SWTID-2026-8331
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	Final Report

1. INTRODUCTION

1.1 Project Overview

The project implements a Script-Controlled Access Control List (ACL) in ServiceNow to restrict record access based on a field value. The demonstrated implementation uses the Institution Details table and a Branch = EEE condition.

1.2 Purpose

The purpose is to provide record-level authorization that combines a field condition with scripted role validation.

2. IDEATION PHASE

2.1 Problem Statement

Security administrators need record-level access decisions that can depend on the value of the requested record rather than only broad role permissions.

2.2 Empathy Map Canvas

The empathy map identifies the Security Administrator / Data Owner as the primary persona and captures what the user thinks and feels, sees, hears, says and does, together with pain and gain areas.

2.3 Brainstorming

The brainstorming stage considered field-based ACL conditions, scripted access checks, role + field validation, Allow/Deny decisions, reusable ACL scripts, audit logging and positive/negative testing.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Discover need → Open access request → Evaluate rule → Receive decision → Review / improve.

3.2 Solution Requirement

Functional requirements cover record access evaluation, access rule management, enforcement and audit, and policy testing. Non-functional requirements cover usability, security, reliability, performance, availability and scalability.

3.3 Data Flow Diagram

User / Role → Access Request → ServiceNow ACL → Field Value + Rule Evaluation → Allow / Deny → Protected Record.

3.4 Technology Stack

ServiceNow UI / Workspace, ServiceNow ACLs, server-side JavaScript, ServiceNow roles, ServiceNow platform database and ServiceNow Cloud.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The field-value-aware ACL directly addresses the gap between broad role permissions and record-specific authorization requirements.

4.2 Proposed Solution

Configure a record-level ACL for the Institution Details table, apply the Branch = EEE condition, and use an Advanced script for role-based authorization.

4.3 Solution Architecture

User → Record Request → ACL Controller → Field Evaluation → Advanced Script → Allow / Deny → Protected Record.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The supplied project-planning template is used with Sprint, Functional Requirement (Epic), User Story Number, User Story / Task, Story Points, Priority and Team Members. Sprint-1 covers access-control setup and evaluation; later sprints cover Allow/Deny, audit and testing.

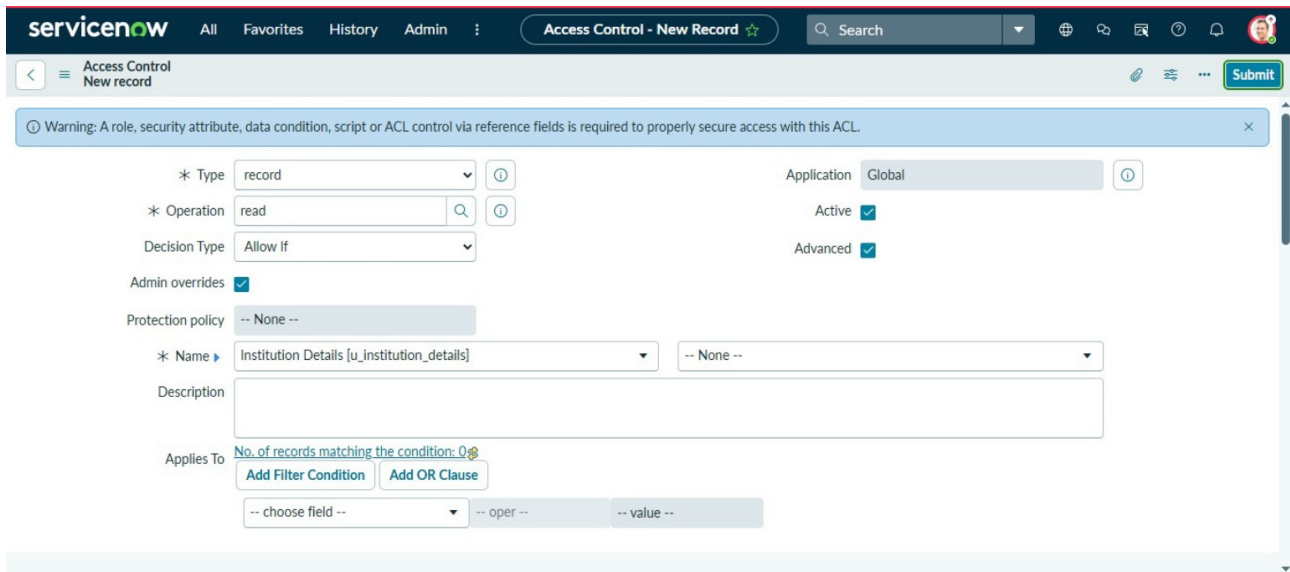
6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project uses deterministic ACL evaluation rather than a machine-learning model. Functional verification therefore focuses on field-value evaluation, the ACL script and the resulting access decision.

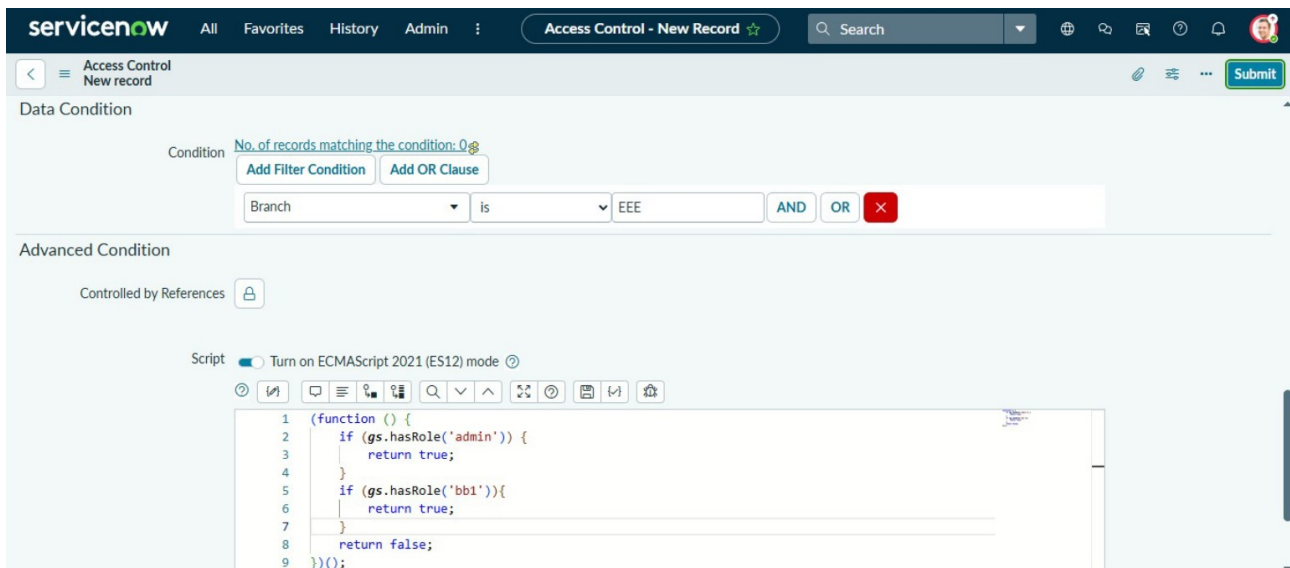
7. RESULTS

7.1 Output Screenshots



The screenshot shows the 'Access Control - New Record' form in ServiceNow. At the top, there's a warning: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' The form includes fields for 'Type' (set to 'record'), 'Operation' (set to 'read'), 'Decision Type' (set to 'Allow If'), 'Application' (set to 'Global'), 'Active' (checked), and 'Advanced' (checked). There's a 'Protection policy' dropdown set to '-- None --'. The 'Name' field is set to 'Institution Details [u_institution_details]'. Below this is a 'Description' field. The 'Applies To' section shows 'No. of records matching the condition: 0' and buttons for 'Add Filter Condition' and 'Add OR Clause'. At the bottom, there are dropdowns for 'choose field --', 'oper --', and 'value --'.

Figure 1. ServiceNow ACL configuration.



The screenshot shows the 'Data Condition' and 'Advanced Condition' sections of the ServiceNow ACL configuration. The 'Data Condition' section has a 'Condition' field set to 'No. of records matching the condition: 0' and buttons for 'Add Filter Condition' and 'Add OR Clause'. Below this is a dropdown for 'Branch' set to 'is', followed by a dropdown for 'EEE' and buttons for 'AND', 'OR', and a red 'X' button. The 'Advanced Condition' section has a 'Controlled by References' checkbox. Below this is a 'Script' section with a toggle for 'Turn on ECMAScript 2021 (ES12) mode' and a code editor. The code editor contains the following JavaScript code:

```
1 (function () {  
2   if (gs.hasRole('admin')) {  
3     return true;  
4   }  
5   if (gs.hasRole('bb1')) {  
6     return true;  
7   }  
8   return false;  
9 })();
```

Figure 2. Branch = EEE condition and Advanced ACL script.

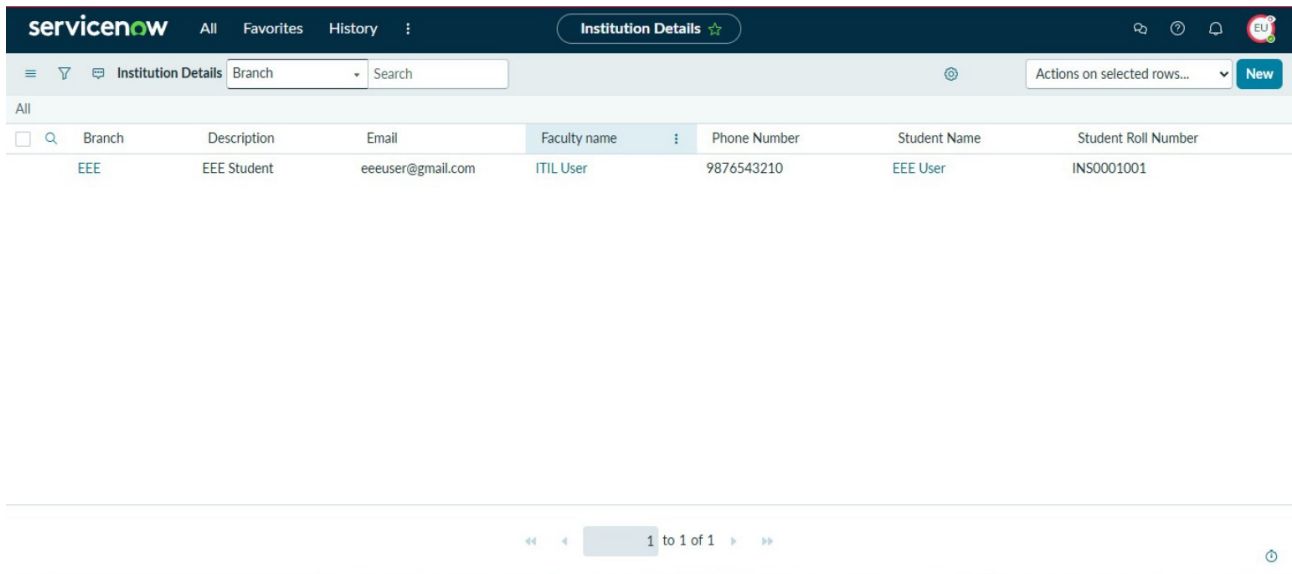


Figure 3. Institution Details record evidence.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Precise record-level control.	Requires careful ACL configuration.
Field values participate in authorization.	Complex scripts may require maintenance.
Role checks can be combined with field conditions.	Negative cases must be tested separately.
Centralized enforcement at the ACL layer.	Incorrect conditions can cause unintended access results.

9. CONCLUSION

The Script-Controlled ACL demonstrates a field-value-based method for restricting ServiceNow record access. The implementation uses the Branch = EEE condition and Advanced authorization logic for the Institution Details record.

10. FUTURE SCOPE

Future scope includes automated regression tests, dedicated denied-access testing, reusable Script Includes, richer audit reporting and support for additional tables and field conditions.

11. APPENDIX

Item	Details
Source Code(if any)	ServiceNow ACL configuration and Advanced server-side script.
Dataset Link	Not applicable to the core ACL implementation.
GitHub & Project Demo Link	Demo Video (Google Drive)